A-A-51206
20 November 1985

COMMERCIAL ITEM DESCRIPTION

BUR, DENTURE TRIMMING,(High Speed Tool Steel)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GG-B-823C.

This commercial item description covers a denture trimming bur used for finishing dentures in prosthetic laboratory procedures.

Salient characteristics:

The bur shall be made of AISI type M2 high speed tool steel. The bur shape shall be as shown in figure 1. The overall length shall be 1-3/4 ± 1/8 inch, the shank diameter shall be 0.0924 + 0.0005 inch, the number of blades shall be 12 ± 1, the head diameter at largest diameter shall be 0.188 ± 0.005 inch and the length of the head shall be 27/64 ± 1/64 inch. The bur shall be a denture trimming bur, suitable for use with a straight handpiece in finishing plastic denture materials in prosthetic laboratory procedures. The bur shall be furnished as specified. The bur shall be of one-piece construction, and shall consist of a shank and a head. The shank of the bur shall be straight. The blades of the bur head shall be shaped and sharpened to cut at the tip as well as the sides of the bur head, and provide automatic chip clearance.

Runout. The total indicated runout of the burs shall not exceed 0.005 inch, when tested as specified in test section under runout.

Hardness. The head and shank of the bur shall have a minimum Rockwell hardness of C62, when tested in accordance with ASTM E-18 (Rockwell) or E-92 (Vickers or Diamond pyramid).

Fatigue in cutting. The time required to cut the eighteenth half-inch shall not be more than 2.25 the time that is required to cut the second half-inch, when tested as specified in test section under fatigue in cutting. The burs shall show no bending as a result of these tests.

Finish. The bur shall have a black oxidized finish. The shank of the bur shall be smoothly finished.

Marking. Marking of the bur with the name or trademark shall be at the option of the supplier.

Workmanship. The bur shall be free from defects which may detract from its appearance and impair its serviceability.

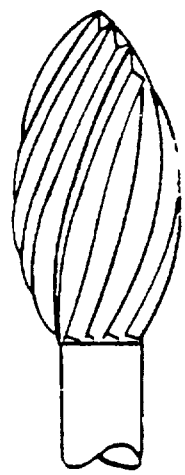
Unit. Package (PG). One package containing twelve burs, as specified, constitutes one unit. The package shall be so designed that each bur is contained in an individual cell.

FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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FLAME

SHAPE OF BUR HEAD

FIGURE 1

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one bur. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests shall be conducted at inspection level S-1, AQL 1.0.

Tests.

Chemical composition. In the absence of certificate of quality for the raw material, the samples selected shall be analyzed in accordance with Method 111 or 112 of FED-STD-151. In case of dispute, Method 111 shall be used for referee analysis.

Runout. The bur shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife edge foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the bur, and so positioned that the foot contacts the bur neck at the point of smallest diameter. The maximum and minimum reading obtained during one complete 360-degree rotation of the bur shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

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Fatigue in cutting.

Side of head. Tests shall be made on a milling machine or similar device. The feed shall be that produced by a 1-pound weight operating a cord attached to a carriage mounted on ball bearings. The bur shall rotate at 1,600 revolutions per minute. The material to be cut shall be 0.020 rolled brass with a Brinell hardness of 74-81 (Rockwell 30-T, 42-49). The edge of the brass shall be centered on the length of the head of the bur. The bur shall be perpendicular to the brass sheet, and the cut shall be made the length of the brass. A recording device operated by a synchronous motor shall be used to time the cut.

Tip of head. Test shall be conducted as in side of head, except that brass test piece shall be applied against the point of the bur.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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NOTE: The following National Stock Number is coverd by this document:

NSN	Item Identification
6520-01-110-3958	BUR, DENTURE TRIMMING, Straight Handpiece, High Speed Tool Steel, 12s

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract or purchase order.

MILITARY INTERESTS:
Custodians:
Army-MD
Navy-MS
Air Force-03

CIVIL AGENCY COORDINATING ACTIVITY:
VA-OSS

PREPARING ACTIVITY:
DLA-DM

Agent:
DoD-MB

49102

Project No. 6520-1732

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